MEZO · MUSD

MUSD Trove Radar + TroveGuard

See liquidations, Recovery Mode and redemptions coming, and let a keeper protect your trove before they hit your BTC.

An open-source radar, alert bot, keeper daemon and on-chain trove guard, live on **Mezo Testnet**

github.com/northbit-dev/musd-trove-radar · northbit-dev.github.io/musd-trove-radar

GPL-3.0

THE PROBLEM

Borrowers can lose their BTC three ways, and nobody shows them where they stand

Liquidation

Below **110 %** collateral ratio the trove is closed and the BTC goes to the Stability Pool.

Recovery Mode

If the system ratio falls below **150 %**, troves under 150 % can be liquidated too.

Redemptions

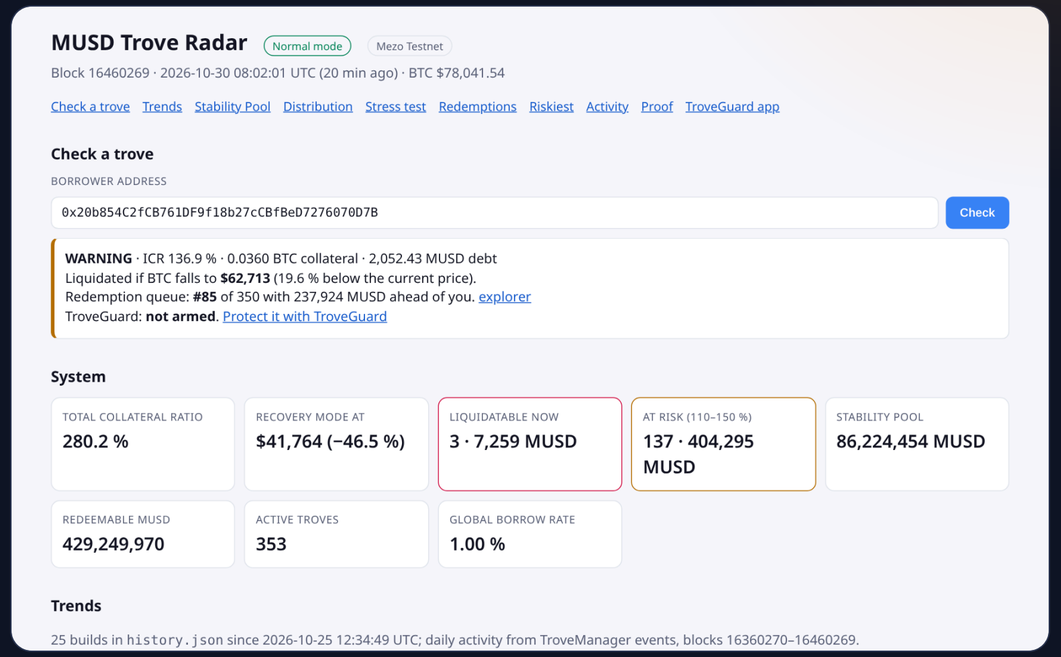
Anyone can swap MUSD for BTC. It is taken from the **lowest-ratio troves first**.

- The Mezo app shows your own trove. It does not show your place among the other 352, or how close the system is to Recovery Mode.
- A warning only helps if you are awake to act on it. Keepers and Stability Pool depositors have no simple view of what is liquidatable and whether it pays.

THE SOLUTION

One tool that reads the MUSD contracts directly, pinned to one block

- **Radar and history:** every trove's ratio, liquidation price, a BTC price-shock stress test, Stability Pool coverage, and trends over time.
- **Redemption queue:** your position, the MUSD ahead of you, and what a redemption of X MUSD would hit.
- **Alerts:** Slack, Discord, Telegram or a signed webhook, on thresholds and when a trove is redeemed or liquidated.
- **TroveGuard:** pre-signed top-ups or repayments that any keeper can execute below your trigger.
- **Keeper daemon:** liquidations and guard `protect` calls, simulated first and sent only when they pay.



LIVE ON MEZO TESTNET · BLOCK 16460269 · BTC $78,042

What the radar sees today

353
active troves

280 %
system collateral ratio

−46.5 %
BTC drop to Recovery Mode ($41,764)

3
liquidatable now, 7,259 MUSD, each checked with `eth_call`

137
troves between 110 and 150 %

92
liquidatable if BTC falls 20 %

11,877×
Stability Pool coverage of liquidatable debt

23
troves a 50,000 MUSD redemption would close

TROVEGUARD V2

Top up or repay automatically, while you sleep

1

Arm

Set a trigger ratio and a keeper tip. Park BTC and/or MUSD in the guard as a reserve.

2

Sign a ladder

One EIP-712 permit per rung, for consecutive nonces: `AddColl` (BTC in) or `RepayMUSD` (debt down).

3

Protect

Below the trigger, anyone calls `protect()`. One rung is spent, funds only enter your own trove, the caller earns the tip.

4

Cancel

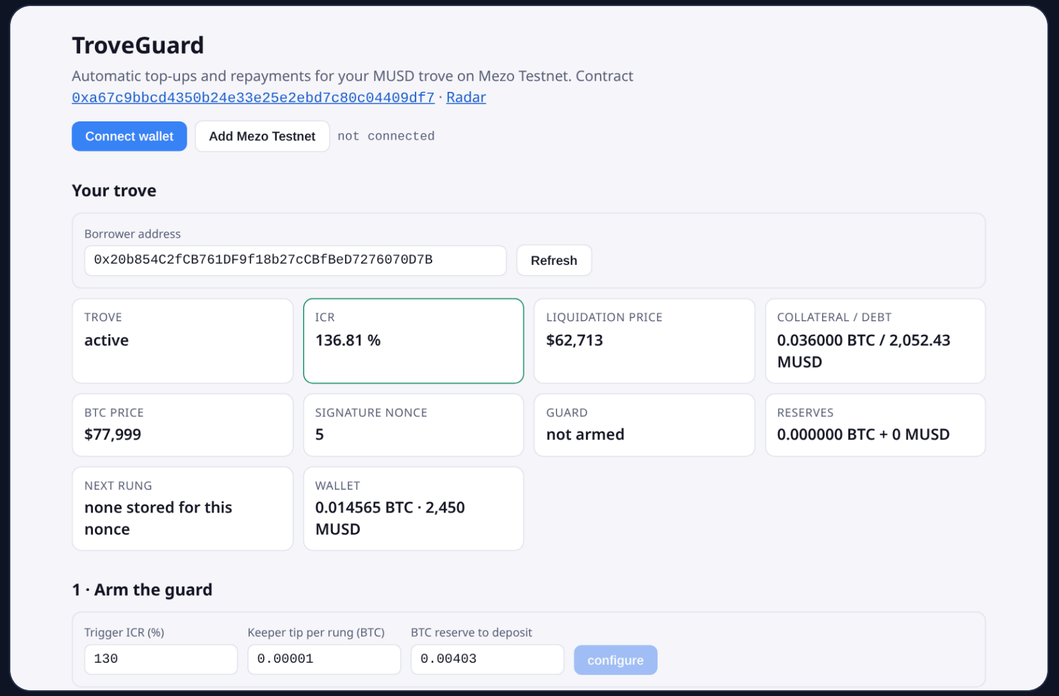
`cancel` refunds every reserve and invalidates every stored permit.

LIVE PROOF, REAL ORACLE, NO MOCKS	RATIO	TRANSACTION
protect, RepayMUSD	118.45 % → 123.95 %	`0x950ca982156cc6b3dd1f6e2e69552e00aa19bba9b70765545b3764947d335aa9`
protect, AddColl	123.88 % → 131.16 %	`0xc3b2dcde9ad22ebd2d24c4aabf87d45b5e72759cc4a615565f53c65cdde6e554`
protect of a permit signed in guard.html	131.23 % → 137.62 %	`0x084a2b4b964c04c722a4d10d32a04d9b8c838d0d9bd4253f7e7edbd47b052212`

Contract 0xa67c9bbcd4350b24e33e25e2ebd7c80c04409df7, verified · 14 fork tests, 3 fuzz tests, 3 invariants · Slither reviewed · SECURITY.md

THE WEB FLOW AND THE KEEPER

A wallet page for borrowers, a daemon for keepers



guard.html

- Any EIP-1193 wallet: switch to Mezo Testnet, see your trove and guard, arm it, fund it, sign the ladder in one `storePermits` tx.
- Every transaction is simulated before the wallet prompt. No keys, nothing loaded from other origins.

Keeper daemon

- `keeper --loop --min-profit 5 --send`: `eth_call` before every send, backoff, JSON logs, Docker image.
- On testnet it liquidated a 3,873 MUSD trove, skipped three below its minimum profit, and recovered from an RPC error by itself.

WHO IT IS FOR

Safer troves mean less bad debt and more trust in MUSD

Borrowers

Early warning, their place in the redemption queue, and a guard that acts for them.

Keepers

An open-source, profit-aware daemon. Liquidations and guard tips are its income.

Stability Pool

Coverage, pool size and expected BTC gains, plus a live per-depositor lookup.

Mezo

More keepers and better-protected troves make MUSD sturdier as BTC-backed money.

How it can pay for itself

- A hosted alert service: free tier, then paid per-trove alerts and Telegram subscriptions.
- An optional protocol fee on guard tips, and a keeper network sharing a hosted RPC and mempool feed.

WHAT'S NEW AND WHAT'S NEXT

Wave 2 shipped. Mainnet is the next step.

Done in Wave 2

- Event history and event alerts
- TroveGuard v2 with repay mode, live on testnet
- Fuzz tests, invariants and a Slither review
- The browser guard page
- Dashboard history and the Stability Pool view
- Keeper daemon and Docker image

Next

- Hosted alerts with per-user subscriptions
- A keeper network: many operators, no duplicate sends
- Mempool alerts for incoming redemptions
- An external audit of TroveGuard
- Mainnet, once MUSD addresses and keeper economics are confirmed

Ask: feedback from the Mezo team on the guard design, and a path to mainnet addresses.

github.com/northbit-dev/musd-trove-radar · northbit-dev.github.io/musd-trove-radar · built by northbit with an AI coding assistant